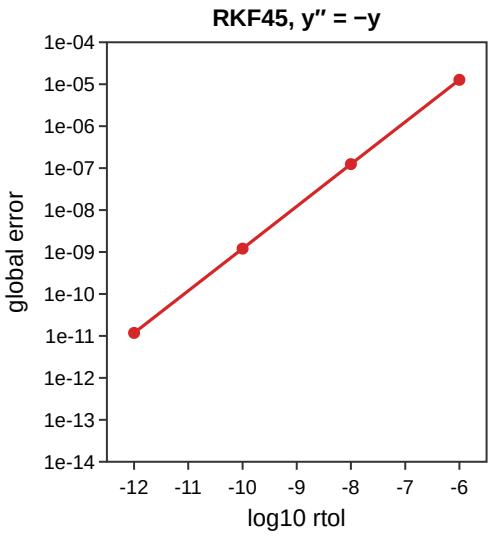
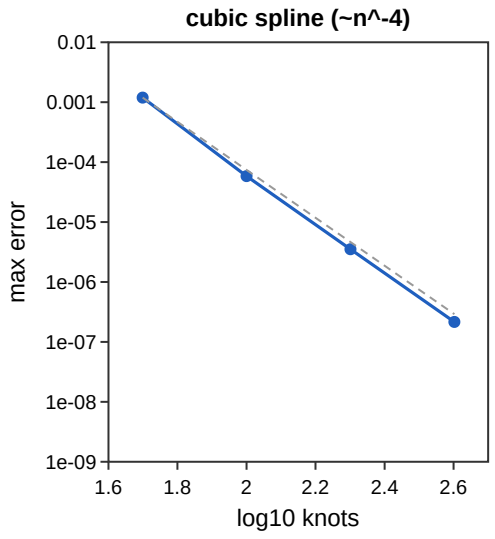
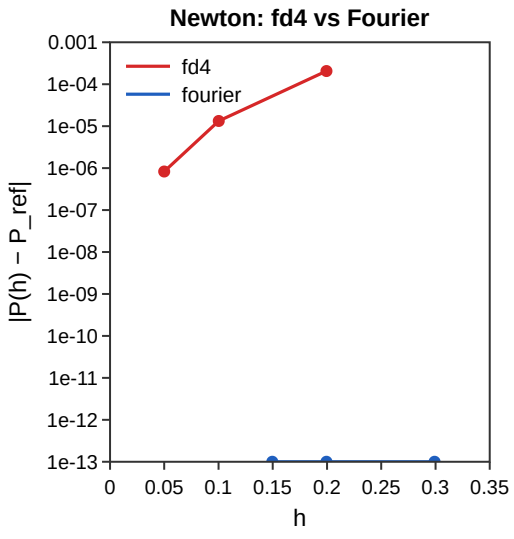
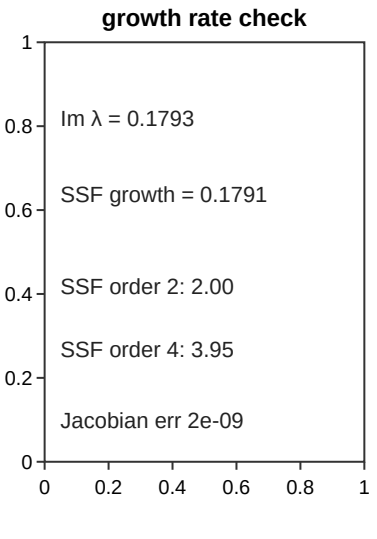

RKF45, y″ = −y
global error
1e-04
1e-05
1e-06
1e-07
1e-08
1e-09
1e-10
1e-11
1e-12
1e-13
1e-14
-12
-11
-10
-9
-8
-7
-6
log10 rtol


cubic spline (~n^-4)
max error
0.01
0.001
1e-04
1e-05
1e-06
1e-07
1e-08
1e-09
1.6
1.8
2
2.2
2.4
2.6
log10 knots


Newton: fd4 vs Fourier
fd4
fourier
|P(h) − P_ref|
0.001
1e-04
1e-05
1e-06
1e-07
1e-08
1e-09
1e-10
1e-11
1e-12
1e-13
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
h


growth rate check
Im λ = 0.1793
SSF growth = 0.1791
SSF order 2: 2.00
SSF order 4: 3.95
Jacobian err 2e-09
1
0.8
0.6
0.4
0.2
0
0
0.2
0.4
0.6
0.8
1